

Othmane Echchabi

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Spatiotemporal machine learning for sustainability systems, with emphasis on remote sensing, climate risk, wildfire modeling, and decision support for environmental monitoring.

EDUCATION

McGill University & Mila – Quebec AI Institute

Montréal, QC

M.Sc. in Computer Science (GPA: 4.0/4.0)

May 2027

Duke University / Duke Kunshan University

Durham, NC / Suzhou, China

B.S. in Data Science

May 2025

- Thesis: *Monitoring Spartina Alterniflora Using Self-Supervised Learning*.
- Study Abroad: Universidad Carlos III de Madrid (Spring 2022).

RESEARCH EXPERIENCE

Doctoral Research & Teaching Assistant

Aug. 2025 – Present

Rolnick Lab, McGill University & Mila

Montréal, QC

- Leading a tree-mortality classification project in collaboration with the Smithsonian Institution on Barro Colorado Island (BCI), Panama; building georeferenced image datasets and developing robust computer-vision pipelines for species-level canopy health assessment across heterogeneous tropical forest conditions.

Machine Learning Research Intern

Jul. 2025 – Dec. 2025

College of Computing, University Mohammed VI Polytechnic

Benguerir, Morocco

- Integrated a 10-year crop-yield dataset with N/P/K fertilizer inputs across 150+ countries for downstream modeling.
- Built a transformer-based recommendation model and a RAG assistant for agronomy; reduced query latency to ~2s.

Undergraduate Research Assistant

Jun. 2024 – Aug. 2025

Sustainability Assessment Group & Chang Lab, Duke Kunshan University

Suzhou, China

- Built a 300k-image, ground-truthed satellite piped-water access dataset and achieved SotA accuracy with SSL foundation models; validated predictions against UN data by running large-scale inference across 50/54 African countries using 1M+ Sentinel images and population-density covariates.
- Built SpeedTransformer for Geolife transport-mode prediction (96% Top-1, +3% vs. baseline), and prototyped CarbonClever, a phone application, to deliver personalized guidance for reducing personal CO₂ emissions.

Undergraduate Research Assistant

Jan. 2025 – Jun. 2025

Malof Lab, University of Missouri

Columbia, MO

- Developed a Perceiver IO-based model and VAE simulator for wildfire spread using satellite and climate data; evaluated on WSTS+ for scenario-driven analyses.

Undergraduate Research Fellow

Jun. 2023 – Aug. 2024

Data+ & Climate+, Duke University

Durham, NC

- Mapped invasive salt-marsh species from Sentinel-2 using ViTs; delivered landscape-scale outputs for conservation planning.
- Built a geospatial database on saltwater intrusion and sea-level rise; delivered an interactive ArcGIS dashboard.

PUBLICATIONS & PREPRINTS

1. Zhang, Y.*; Echchabi, O.*; Feng, T.; Zhang, W.; Liao, H.-K.; Chang, C. *Detecting Transportation Mode Using Dense Smartphone GPS Trajectories and Transformer Models*. Accepted at *International Journal of Geographical Information Science* (2026).
2. Echchabi, O.*; Lahlou, A.*; Talty, N.; Manto, J. M.; Tongshu, Z.; Lam, K. L. *Monitoring Access to Piped Water and Sanitation Infrastructure in Africa at Disaggregated Scales Using Satellite Imagery and Self-Supervised Learning*. Accepted at the *Machine Learning for Remote Sensing Workshop, International Conference on Learning Representations* (2026).
3. Echchabi, O.; Lahlou, A.; Talty, N.; Manto, J. M.; Lam, K. L. *Tracking Progress Towards Sustainable Development Goal 6*

Using Satellite Imagery. arXiv preprint.

*Equal contribution.

POSTERS & PRESENTATIONS

- Monitoring *Spartina alterniflora* Using Self-Supervised Learning. *Duke Climate+ Symposium* (2024).
- Assessing Climate Change Risk of Rural Coastal Plains. *Duke Climate+ Symposium* (2023).
- Lahlou, A.; Echchabi, O.; Talty, N.; Manto, J. M.; Tongshu, Z.; Lam, K. L. *Scalable Water and Sanitation Monitoring Across Africa via AI-Driven Satellite and Survey Data Integration*. *American Geophysical Union Annual Meeting* (2025).

GRANTS & AWARDS

- McGill CS Graduate Award — \$30,000/year (2025)
- McGill CS Graduate Registration Bonus — \$5,000 (2025)
- DKU Student Experiential Learning Fellowship — \$1,000 (2024)
- DKU Signature Work Research Grant — \$1,500 (2023)
- DKU Innovation Incubator — \$3,000 (2023)

TEACHING EXPERIENCE

- **McGill University** — Teaching Assistant (2025–Present) — COMP 303;
- **Duke Kunshan University** — Math & CS Teaching Assistant (2021–2025); Courses: COMPSCI 101, COMPSCI 201, MATH 105, MATH 201, MATH 202, MATH 317;
- **FADI Academy - Online College Prep Center** — Partner & Math Tutor

SERVICE & LEADERSHIP

Mila - AI Quebec Institute

2025 – Present

Mila Lab Representative

Montréal, QC

- Elected as 1 of 11 Mila Lab Representatives, representing 900+ students across Mila programs.
- Active member of AEM (*Association Etudiante du Mila*) and contributor to student-facing initiatives.

McGill University

2025 – Present

Project Development Vice President (Board Member), Muslims in Tech

Montréal, QC

- Coordinate community initiatives and technical programming for student and early-career members.

Duke Kunshan University

2021 – 2025

Resident Assistant; Soccer Varsity Co-Captain; CS Club Software Tech Lead

Suzhou, China

- As Resident Assistant, organized community-led, fully student-run courtyard events with 300+ attendance on a limited budget, and launched recurring traditions that continue under subsequent RA cohorts (e.g., The Midnight Breakfast).
- As CS Club Software Tech Lead, organized and chaired DKU's largest hackathon to date, leading end-to-end planning across sponsorship, logistics, judging, and participant support; also led hands-on technical workshops to prepare students for competitive project building.
- As Soccer Varsity Co-Captain, led the team to 1st place in the Sino-Foreign University Cup and Schwarzman Cup, and 4th place in the Suzhou League.

FIFA World Cup 2022

2022

Team Leader, Spectator Services Volunteer

Doha, Qatar

- Assigned to two 40,000+ seat stadiums, supported high-volume fan flows, resolved on-site spectator issues, and coordinated volunteer workflows to help ensure smooth event operations throughout the tournament.

MISCELLANEOUS

Technical Skills : Python, PyTorch, Docker, Kubernetes, OpenCV, Linux, Git, TensorFlow, Keras, scikit-learn, Java

Spoken Languages : Fluent in Arabic, French, English; Intermediate Spanish and Chinese

Interests : Competitive Soccer, Weightlifting, Cooking, Traveling.